FRTN30 Assignment 3

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1.1 a)

The simulation gave an average return time of 6.733 (from a to a).

1.2 b)

Using theory from lectures notes ch. 7.5, we know that the continuous-time Markov chain satisfies

$$\frac{d}{dt}\pi(t) = -L^T\pi(t), \qquad L = \text{diag}(\omega) - \Lambda,$$

where $\omega_i = \sum_j \Lambda_{ij}$, and that for a stationary probability vector π the following holds,

$$L^T\pi = 0, \qquad \mathbf{1}^T\pi = 1.$$

Solving the linear system gives

$$\pi = \left(\tfrac{5}{27}, \tfrac{4}{27}, \tfrac{2}{9}, \tfrac{2}{9}, \tfrac{2}{9}\right), \quad \Rightarrow \quad \pi_a = \tfrac{4}{27}.$$

The mean return time to state a is computed by,

$$\mathbb{E}_a[\bar{T}_a^+] = \frac{1}{\pi_a\,\omega_a},$$

where $\omega_a = \sum_j \Lambda_{aj} = 1$.

Then we get

$$\mathbb{E}_a[\bar{T}_a^+] = \frac{1}{4/27} = \frac{27}{4} = 6.75$$

Which corresponds to our simulated result.

1.3 c)

According to the simulations, the average hitting time from node "o" to node "d" is 8.78.

1.4 d)

To calculate the theoretical hitting time we use the definition in the lecture notes:

$$E_i[T_d] = \frac{1}{w_i} + \sum_j P_{i,j} E_j[T_d], i \notin d \tag{1}$$

We also know that $E_d[T_d] = 0$ since the expected time to hit state d, starting in state d obviously is 0. w_i is the leaving rate for a certain state i and $P_{i,j}$ is the normalized transition matrix $\frac{\Lambda_{i,j}}{\omega_i}$. From this we can calculate the expected hitting time with respect to the graph:

$$\begin{aligned} E_o &= \frac{5}{3} + \frac{2}{3}E_a + \frac{1}{3}E_b \\ E_a &= 1 + \frac{3}{4}E_b + \frac{1}{4}E_c \\ E_b &= 1 + \frac{1}{2}E_o + \frac{1}{2}E_c \\ E_c &= 1 + \frac{1}{3}E_b + \frac{2}{3}E_d = 1 + \frac{1}{3}E_b \end{aligned} \tag{2}$$

When solving this system of equation the answer become:

$$E_o[T_d] = \frac{123}{14} = 8.785 \tag{3}$$

Hence, the simulated answer is fairly similar to the theoretical and we can approve our simulation-technique.

2

2.1 a

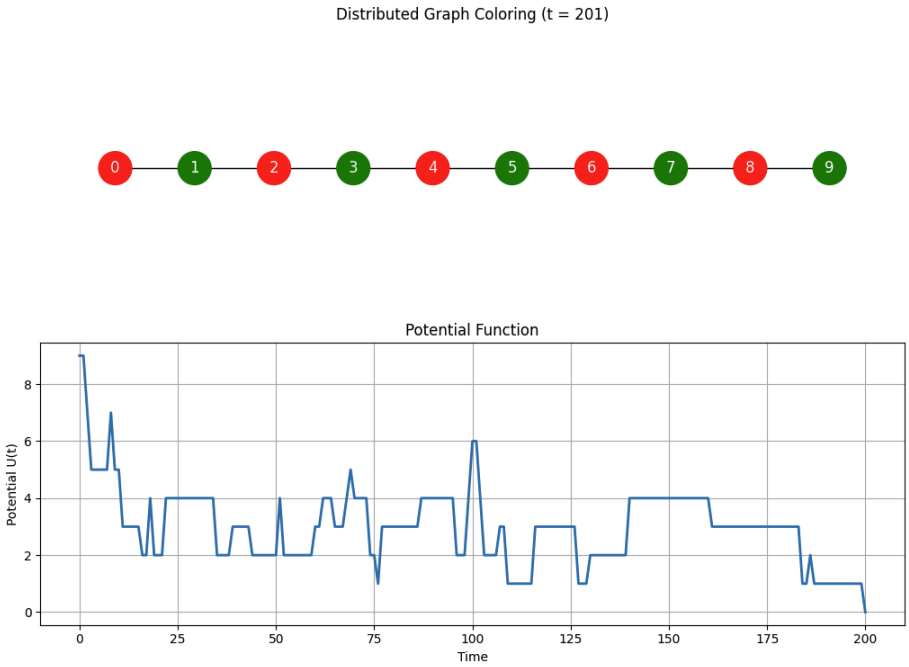


Figure 1: Plot of the potential function as well as final configuration.

As we see in the graph when we start with all red we get a potential function value of 9 which represents the number of clashing neighbors, then we see a decreasing trend until we get a potential of 0 and the algorithm terminates. Sometimes the potential increases due to the stochasticity of the color updating rule, which is noticed more for small values of t when the noise is greater, but with increasing t the noise decreases.

2.2 b

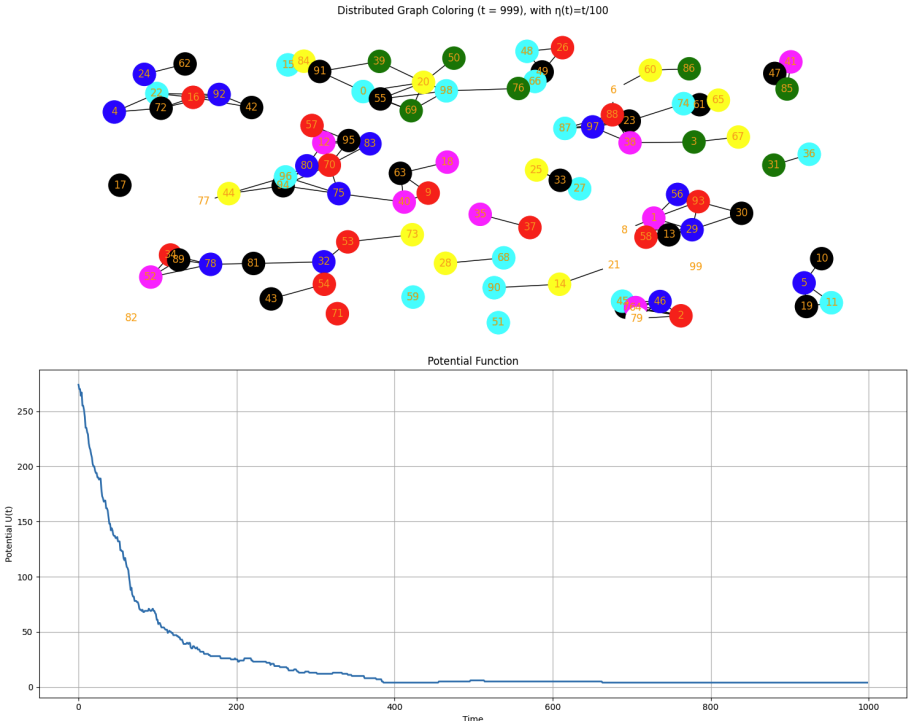


Figure 2: Plot of the potential function as well as final configuration. $\eta = t/100$

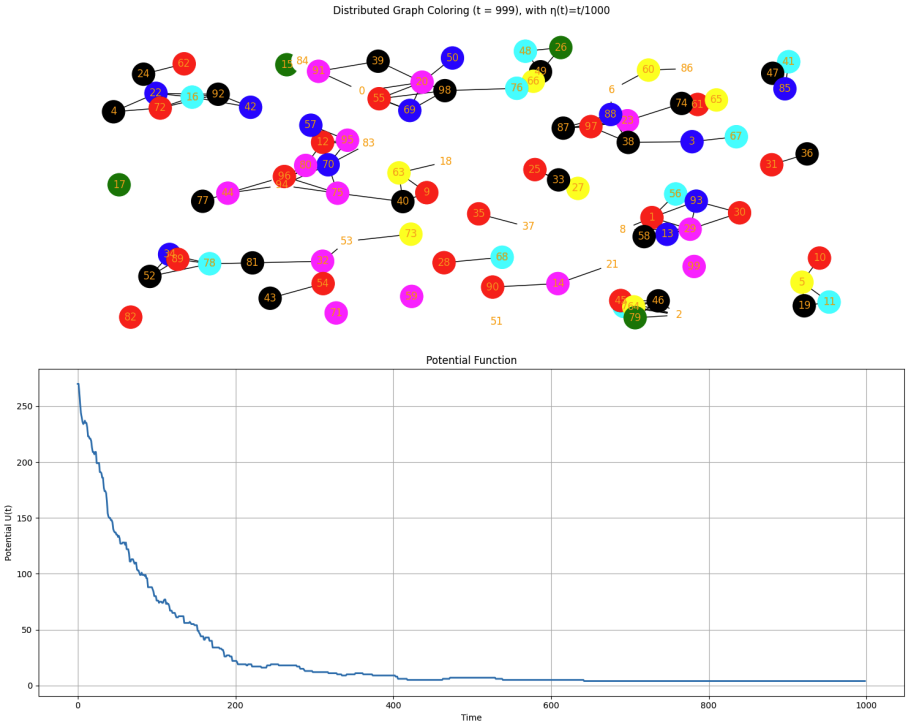


Figure 3: Plot of the potential function as well as final configuration. $\eta = t/1000$

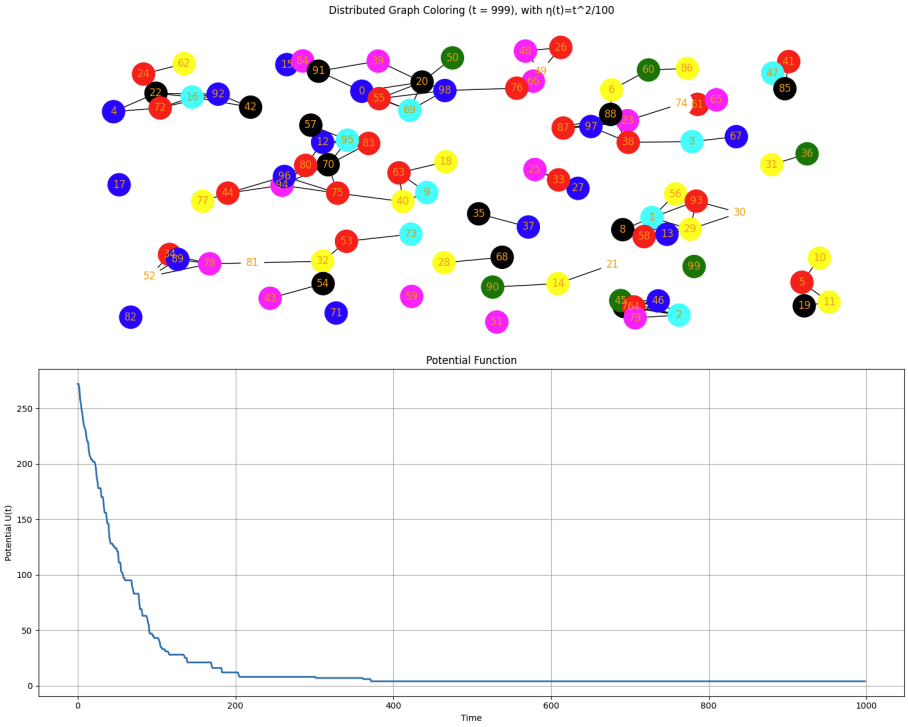


Figure 4: Plot of the potential function as well as final configuration. $\eta = t^2/100$

In Figure 2-4 the model converge towards a near zero potential. When comparing the value for η for a certain t, a larger value gives a faster convergence rate, but with less exploration. Hence, the model is more likely to get stuck in a local minima. With a slowly increasing $\eta(t)$ the system maintain a higher level of randomness for a longer time and more possible minimas are explored. The choice of $\eta(t)$ is a trade of between exploration and exploitation.